

DSML-301 and ASD-501 trend towards the upper end of that range.

SEF-101 — Software Engineering Foundations

Duration: 9 months Level: Beginner Cohorts: Monthly

List tuition: INR 2,10,000 exclusive of GST

SEF-101 is our entry program and the only Meridian program that requires no degree and no prior coding experience. It is designed for career changers, non-engineering graduates, and Class 12 completers who want a first structured route into software engineering.

Syllabus

Module 1 — Programming Foundations (6 weeks)

Python syntax and semantics, control flow, functions, error handling, file I/O, modules and packages, virtual environments, and an introduction to version control with Git.

Module 2 — Data Structures (7 weeks)

Arrays, strings, linked lists, stacks, queues, hash tables, trees, heaps, and graphs. Emphasis on implementation from first principles before use of library types.

Module 3 — Algorithms and Complexity (7 weeks)

Asymptotic analysis, sorting and searching, recursion, two pointers, sliding window, binary search on answer, greedy methods, dynamic programming, and graph traversal.

Module 4 — Databases and SQL (5 weeks)

Relational modelling, normalisation, SQL querying and joins, indexing, transactions, and an introduction to non-relational stores.

Module 5 — Backend Web Development (7 weeks)

HTTP, REST API design, request handling, authentication and authorisation, ORMs, caching basics, and deploying a service.

Module 6 — Engineering Practice (4 weeks)

Testing, debugging, code review, CI basics, reading unfamiliar codebases, and an introduction to system design at interview level.

Capstone projects

Two required. Project A is a command-line application with persistent storage. Project B is a deployed REST API with authentication, tests, and a written design note.

Career outcomes

Typical target roles are Junior Software Engineer, Backend Engineer (entry), and Software Development Engineer I. SEF-101 includes full placement support services. SEF-101 is not eligible for the Placement Assurance conditional refund product.

FSWD-201 — Full-Stack Web Development

Duration: 11 months | Level: Intermediate | Cohorts: Monthly

List tuition: INR 2,85,000 exclusive of GST

FSWD-201 takes learners who already have some coding exposure and builds production-grade full-stack capability. It requires a bachelor's degree (or final-year enrolment) and at least six months of demonstrable coding exposure.

Syllabus

Module 1 — Modern JavaScript and TypeScript (6 weeks)

ES2023 language features, asynchronous programming, modules, the type system, generics, and tooling.

Module 2 — Frontend Engineering with React (8 weeks)

Component architecture, hooks, state management, routing, forms, data fetching, accessibility, and performance profiling.

Module 3 — Backend Services with Node (8 weeks)

Express and Fastify, middleware, validation, authentication with JWT and sessions, file handling, background jobs, and WebSockets.

Module 4 — Data Layer (6 weeks)

PostgreSQL modelling and query tuning, Redis for caching and queues, migrations, connection pooling, and an introduction to MongoDB.

Module 5 — System Design for Web Applications (7 weeks)

Load balancing, horizontal scaling, caching strategies, rate limiting, idempotency, eventual consistency, and observability.

Module 6 — Delivery and Operations (6 weeks)

Docker, CI/CD pipelines, deployment on cloud platforms, monitoring, logging, incident response, and cost awareness.

Capstone projects

Three required. A production-quality full-stack application, a public API with documented contracts and rate limiting, and a performance optimisation case study on a supplied slow codebase.

Career outcomes

Typical target roles are Full-Stack Engineer, Frontend Engineer, and Backend Engineer. FSWD-201 includes full placement support and is one of only two programs eligible for Placement Assurance, with a minimum assured CTC of INR 7,00,000.

DSML-301 — Data Science and Machine Learning

Duration: 13 months | Level: Advanced | Cohorts: Every two months

List tuition: INR 3,40,000 exclusive of GST

DSML-301 is our longest and most demanding program. It requires a bachelor's degree with a quantitative component, at least twelve months of full-time work experience, and a technical screening call. Learners receive USD 400 of cloud GPU credits for model training; no local GPU is required.

Syllabus

Module 1 — Mathematics for Machine Learning (7 weeks)

Linear algebra, calculus for optimisation, probability, and statistical inference including hypothesis testing and confidence intervals.

Module 2 — Data Engineering and Analysis (8 weeks)

Python data stack, SQL at analytical scale, data cleaning, feature engineering, exploratory analysis, and visualisation.

Module 3 — Classical Machine Learning (9 weeks)

Linear and logistic regression, regularisation, decision trees, random forests, gradient boosting, support vector machines, clustering, dimensionality reduction, and rigorous model evaluation.

Module 4 — Deep Learning (9 weeks)

Neural network fundamentals, backpropagation, convolutional networks for vision, recurrent and attention-based architectures for sequence data, transfer learning, and training at scale.

Module 5 — Natural Language Processing and Large Language Models (8 weeks)

Text representation, transformer architectures, fine-tuning, prompt engineering, retrieval augmented generation, evaluation of generative systems, and responsible deployment.

Module 6 — Machine Learning in Production (7 weeks)

Experiment tracking, model packaging and serving, feature stores, monitoring and drift detection, A/B testing, and cost and latency management.

Capstone projects

Three required. An end-to-end supervised learning project with a written evaluation report, a deep learning project on vision or language, and a deployed inference service with monitoring.

Career outcomes

Typical target roles are Data Scientist, Machine Learning Engineer, and Applied Scientist (entry to mid). DSML-301 includes full placement support and is eligible for Placement Assurance, with a minimum assured CTC of INR 7,00,000.

DCE-401 — DevOps and Cloud Engineering

Duration: 8 months | Level: Intermediate | Cohorts: Every two months

List tuition: INR 2,45,000 exclusive of GST

DCE-401 is for professionals already working in an IT role who want to move into infrastructure, platform, or reliability engineering. It requires twelve months of IT experience and comfort with the Linux command line. Learners receive USD 250 of cloud lab credits.

Syllabus

Module 1 — Linux and Networking Foundations (5 weeks)

Process and memory management, filesystems, permissions, shell scripting, TCP/IP, DNS, HTTP, TLS, and troubleshooting methodology.

Module 2 — Infrastructure as Code (5 weeks)

Terraform, state management, modules, drift detection, and configuration management with Ansible.

Module 3 — Containers and Orchestration (7 weeks)

Docker internals, image optimisation, Kubernetes architecture, workloads, services, ingress, storage, autoscaling, and Helm.

Module 4 — CI/CD and Release Engineering (5 weeks)

Pipeline design, artefact management, testing gates, blue-green and canary deployment, feature flags, and rollback strategy.

Module 5 — Observability and Reliability (6 weeks)

Metrics, logs, traces, service level objectives, error budgets, alerting design, on-call practice, and incident postmortems.

Module 6 — Cloud Architecture and Security (6 weeks)

Compute and storage services, networking and VPC design, identity and access management, secrets handling, cost optimisation, and compliance basics.

Capstone projects

Two required. A fully automated multi-environment infrastructure deployment, and an observability and incident-response exercise on a deliberately fragile system.

Career outcomes

Typical target roles are DevOps Engineer, Site Reliability Engineer, Cloud Engineer, and Platform

Engineer. DCE-401 includes full placement support. DCE-401 is not eligible for Placement Assurance.

ASD-501 — Advanced Systems Design

Duration: 6 months | **Level:** Senior | **Cohorts:** Quarterly (Jan, Apr, Jul, Oct)

List tuition: INR 1,95,000 exclusive of GST

ASD-501 is a senior specialisation for engineers with at least four years of professional software engineering experience, of which at least two years must involve building or operating production backend systems. The Meridian Aptitude Test is waived; admission is decided by a 45-minute technical interview with a senior instructor. Approximately 40% of applicants are declined at that stage.

ASD-501 is explicitly not a career-switching program. It includes no placement services of any kind and is not eligible for Placement Assurance. Applicants who need placement support should choose a different program.

Syllabus

Module 1 — Foundations of Distributed Systems (4 weeks)

Failure models, consistency models, the CAP and PACELC framings, consensus, clocks and ordering, and quorum systems.

Module 2 — Data at Scale (5 weeks)

Storage engine internals, replication and partitioning strategies, distributed transactions, change data capture, and polyglot persistence.

Module 3 — Event-Driven Architecture (4 weeks)

Log-based messaging, exactly-once semantics in practice, stream processing, event sourcing, CQRS, and the saga pattern.

Module 4 — Performance Engineering (4 weeks)

Latency budgets, queueing theory for engineers, profiling, load and stress testing, capacity planning, and back-of-envelope estimation.

Module 5 — Resilience and Operability (4 weeks)

Timeouts, retries and backoff, circuit breakers, bulkheads, graceful degradation, chaos experiments, and designing for on-call.

Module 6 — Architecture in Practice (5 weeks)

Trade-off analysis, architecture decision records, migration and strangler patterns, cost and organisational constraints, and leading a design review.

Capstone projects

Two required. A full design document for a large-scale system with an explicit trade-off analysis, and a live design review presented to and critiqued by a panel of senior instructors.

Career outcomes

ASD-501 is intended to support internal progression to Senior Engineer, Staff Engineer, and Architect roles. Meridian Academy provides no placement services for this program.

Instructors

Meridian instructors are practising engineers, not full-time academics. Every instructor has at least six years of industry experience and passes an internal teaching evaluation before leading a cohort. Instructors are assigned per module, so learners are taught each subject by a specialist rather than by a single generalist across the whole program.

Mentor groups of at most 12 learners meet weekly. Mentors are separate from instructors and are responsible for progress tracking, project feedback, and escalation of learners who fall behind.

Certification

On completion of all six module assessments, the required capstone projects, and the final technical evaluation, learners receive a Meridian Academy Certificate of Completion bearing a verifiable credential ID. Certificates are issued within 21 days of program completion. A reissue costs INR 1,500.

The certificate is an industry certificate. It is not a degree and is not accredited by the UGC or AICTE.

Next steps

Applications are made at the Meridian Academy website. Admission requires the Meridian Aptitude Test, and for DSML-301, DCE-401, and ASD-501 a technical screening call. Full requirements are in the Eligibility Criteria document, MA-ELIG-2026-01. Pricing, discounts, and financing terms are in the pricing master, MA-PRICING-2026-03. Placement services and the Placement Assurance product are described in the Placement Policy, MA-PLACE-2026-02.